

Nimillux: A Locally-Deployed, Verification-First RAG Architecture with Three-Tier Evidential Confidence Estimation

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Abstract. We present Nimillux, a fully local retrieval-augmented generation (RAG) system for question answering over large document corpora, architected around a verification-first design principle in which every generated response undergoes source-grounded claim verification prior to being surfaced. Unlike conventional RAG deployments, Nimillux imposes a hard zero-dependency constraint on commercial inference APIs, eliminating both third-party data exposure and provider-side content-moderation policy as confounding variables in system behavior. We demonstrate empirically that naive binary claim classification systematically masks a substantial population of weakly-supported claims, reaching 60% of claims under cross-domain querying, that are indistinguishable from strongly-supported claims under a binary scheme. We introduce a three-tier evidential confidence model that recovers this latent signal, together with an architecturally-independent secondary verification pathway, and validate system robustness against a hand-labeled evaluation suite that grew from 6 to 27+ entries over the course of development. We further report an empirically-derived hardware operating envelope for consumer-grade local deployment, and characterize ten distinct categories of query-understanding failure, nine closed and protected under regression testing and one remaining open.

Keywords: retrieval-augmented generation, hallucination mitigation, claim verification, confidence calibration, local LLM deployment

1 Introduction

An unverified language-model output presents a specific epistemic risk in research contexts: a fluent, confidently stated generation is not distinguishable, under surface inspection, from one that is faithfully grounded in the underlying corpus. This work is motivated by the position that an unverifiable response constitutes negative value relative to no response at all, and that claim verification must therefore be treated as a first-class architectural primitive rather than a post-hoc auxiliary check.

A second, non-negotiable design constraint applies throughout the system: zero reliance on commercial inference APIs. This constraint is motivated jointly by (i) data-confidentiality requirements over the source corpus, and (ii) elimination of provider-side content-moderation policy as an uncontrolled confound in system behavior.

1.1 Contributions

This report makes the following contributions:

1. A verification-first RAG architecture with zero external API dependency, suitable for privacy-sensitive research corpora.
2. Empirical evidence that binary claim-support classification systematically conflates weak and strong evidential support, with the effect most pronounced under cross-domain querying (Section 6).

3. A three-tier evidential confidence framework that recovers the masked signal, together with an architecturally-independent secondary verification pathway.
4. A regression-tested characterization of ten distinct query-parsing failure modes: nine resolved and regression-protected, one open.
5. An empirically-derived hardware operating envelope for consumer-grade local deployment.

2 System Architecture

Nimillux implements a standard RAG pipeline (Lewis et al., 2020) augmented with a dedicated verification layer (Figure 1):

- **Ingestion:** PDFs processed via OCR for scanned material, segmented under a semantic chunking policy.
- **Indexing:** Chunks embedded into a FAISS vector store (Johnson et al., 2019), supplemented with BM25 lexical retrieval (Robertson & Zaragoza, 2009) and a maintained concept registry that canonicalizes query terms and resolves known aliases.
- **Generation:** A locally-hosted LLM (Mistral 7B; Jiang et al., 2023) produces responses conditioned on retrieved passages.

- **Verification:** Each response is subjected to multi-method source-grounded claim verification prior to being surfaced (Section 4).
- **Interface:** Multi-turn conversational state, cross-document synthesis, and a dedicated contradiction-detection mode.

Dense retrieval selects the top- k corpus chunks by embedding similarity,

$$\text{Top}_k(q) = \arg \max_{d \in \mathcal{D}} \text{sim}(q, d),$$

and is supplemented by lexical scoring under Okapi BM25 (Robertson & Zaragoza, 2009),

$$\text{BM25}(q, d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{f(t, d)(k_1 + 1)}{f(t, d) + k_1 \left(1 - b + b \frac{|d|}{\text{avgl}}\right)},$$

which additionally drives query-term expansion against the concept registry prior to dense retrieval, where $f(t, d)$ is the term frequency of t in chunk d , $|d|$ the chunk length, avgl the corpus-average chunk length, and k_1, b the standard BM25 hyperparameters.

3 Problem Formulation

Let c denote an atomic claim extracted from a generated response, and let $S = \{s_1, \dots, s_k\}$ denote the set of retrieved candidate source passages. We define the verification problem as defining a classifier

$$f : (c, S) \longrightarrow \{\text{strong, weak, unsupported}\}$$

that reflects the degree of evidential support the best-aligned passage $s^* = \arg \max_{s \in S} \text{sim}(c, s)$ provides for c , where sim is embedding cosine similarity:

$$\text{sim}(c, s) = \frac{\mathbf{v}_c \cdot \mathbf{v}_s}{\|\mathbf{v}_c\| \|\mathbf{v}_s\|}.$$

3.1 On Calibration

A scoring function is well-calibrated if, for every predicted confidence level p , $\Pr(y = 1 \mid \hat{p} = p) = p$ (Platt, 1999; Zadrozny & Elkan, 2002). The classifier f defined below is a threshold-based approximation to this ideal: it partitions $\text{sim}(c, s^*)$ into ordinal confidence bands via fixed cut points, rather than fitting a calibration function against ground-truth correctness labels. We use “calibrated” advisedly: tier membership is empirically associated with support quality (Section 6), but full probabilistic calibration via isotonic or Platt-scaled regression against the golden suite remains future work (Section 10).

Algorithm 1 Three-Tier Claim Verification

Require: claim c , candidate passages S , thresholds $\theta_{\text{strong}} = 0.65, \tau = 0.55$
Ensure: classification $\ell \in \{\text{strong, weak, unsupported}\}$
1: $s^* \leftarrow \arg \max_{s \in S} \text{sim}(c, s)$
2: $\sigma \leftarrow \text{sim}(c, s^*)$
3: **if** $\sigma \geq \theta_{\text{strong}}$ **then**
4: $\ell \leftarrow \text{strong}$
5: **else if** $\sigma \geq \tau$ **then**
6: $\ell \leftarrow \text{weak}$
7: **else**
8: $\ell \leftarrow \text{unsupported}$
9: **end if**
10: **return** ℓ

4 Verification Methodology

The initial instantiation of f collapsed the codomain to a binary classification, $\{\text{supported, unsupported}\}$, thresholded on $\text{sim}(c, s^*)$. Output under this scheme appeared uniformly clean: entries frequently reported “100% supported.”

A post-hoc audit of the underlying per-claim similarity scores revealed that the binary threshold was systematically obscuring a latent distribution: on the evaluation suite, every entry previously reported at 100% support in fact contained at least one claim scoring in the 0.55–0.65 interval, a borderline evidential population rendered invisible by the binary formulation. This constitutes a false-precision failure mode: the system was not wrong about *whether* a claim was supported, but was systematically overconfident about *how well*.

We address this by redefining f as a three-tier evidential classifier (Algorithm 1):

$$f(c, S) = \begin{cases} \text{strong} & \text{sim}(c, s^*) \geq 0.65 \\ \text{weak} & 0.55 \leq \text{sim}(c, s^*) < 0.65 \\ \text{unsupported} & \text{sim}(c, s^*) < 0.55 \end{cases}$$

Empirical results from this redesign are reported in Section 6.

5 Evaluation Infrastructure

System behavior is validated against a hand-labeled golden query suite that grew from 6 to 27+ entries over the course of development (Figure 2), each entry specifying expected retrieved concepts, minimum coverage thresholds, and expected routing domain where applicable. No candidate change is considered complete until validated against the full suite without regression.

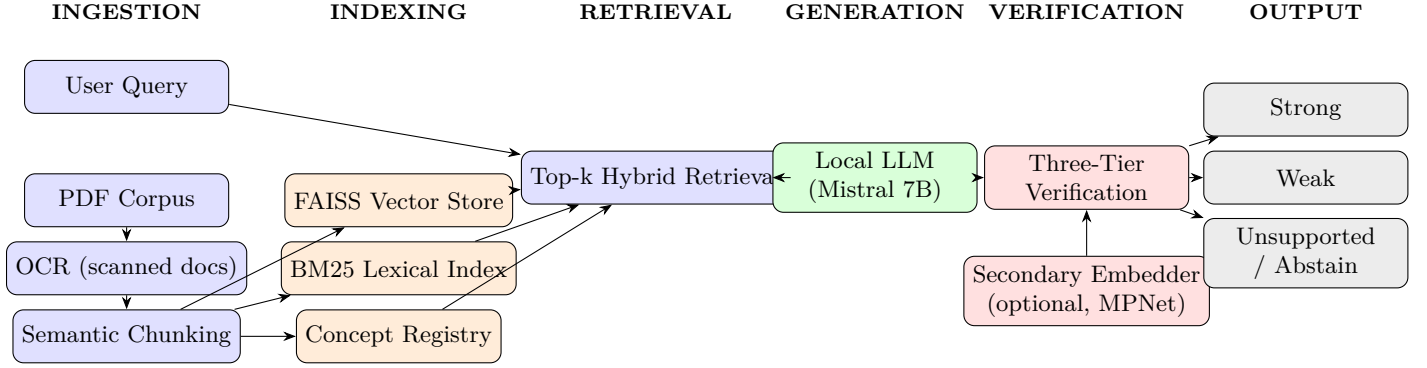


Figure 1: End-to-end system pipeline. Query flow (top) joins the corpus flow (bottom) at retrieval; the secondary embedder participates only in verification, not retrieval.

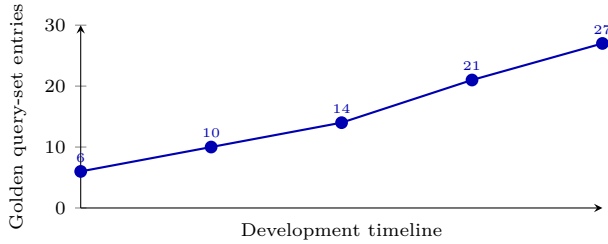


Figure 2: Growth of the hand-labeled evaluation suite over development. Treated as a living regression benchmark rather than a static one.

Table 1: Per-claim grounding results under the three-tier classifier. A sixth entry, **missing-sentinel-arthistory**, triggers the coverage-zero abstention gate prior to generation and is excluded from this table by design.

Entry	Strong	Weak	Weak %
hilbert-vn-algebra	6	1	14%
diffforms-derham-chern	10	1	9%
spin-dirac-atiyahsinger	7	1	13%
compound-diffeom-six	5	4	44%
cross-domain-ml-geometry	2	3	60%

Each entry defines a coverage metric

$$\text{Coverage}(q) = \frac{|\mathcal{C}_{\text{ret}}(q) \cap \mathcal{C}_{\text{exp}}(q)|}{|\mathcal{C}_{\text{exp}}(q)|},$$

where $\mathcal{C}_{\text{exp}}(q)$ is the hand-labeled concept set a query is expected to anchor and $\mathcal{C}_{\text{ret}}(q)$ the concept set actually retrieved; a change is accepted only if $\text{Coverage}(q)$ meets a per-entry minimum locked at authoring time.

This infrastructure has demonstrated concrete defect-detection capability during development rather than post-hoc. An early fuzzy-matching implementation for typo tolerance, employing unordered token-overlap scoring, was rejected after producing a false-positive match on an out-of-domain query in the evaluation suite: direct evidence that naive lexical fuzzy-matching is insufficiently discriminative for this task. The implementation

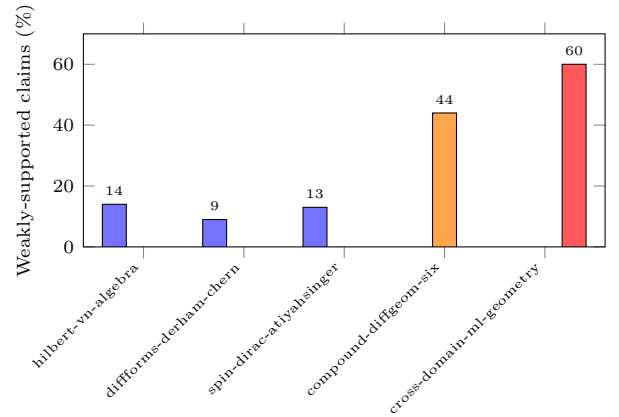


Figure 3: Weak-claim rate by query taxonomy. Single-domain queries cluster below 15%; compound and cross-domain queries exhibit a more than four-fold increase.

was revised to enforce sliding-window token adjacency:

$$\text{ratio}(a, b) = \frac{2M}{T}, \quad \text{accept if } \text{ratio}(a, b) \geq 0.72,$$

where M is the number of matching elements between token sequences a and b under longest-matching-subsequence alignment and T is the total token count across both sequences. This preserved typo tolerance while eliminating the false-positive class observed under the unordered scheme, confirmed against the regression suite.

6 Results

We define the weak-claim rate for query q as

$$\text{WeakRate}(q) = \frac{|\{c : f(c, S) = \text{weak}\}|}{|\{c : f(c, S) \in \{\text{strong}, \text{weak}\}\}|},$$

the fraction of graded (non-abstained) claims falling below the strong-support threshold θ_{strong} . Table 1

reports the per-claim grounding distribution obtained from a representative evaluation sweep under the three-tier classifier; Figure 3 aggregates WeakRate by query taxonomy (single-domain, compound, cross-domain).

The central empirical finding is a strong positive association between query compositionality and claim-level uncertainty: single-domain queries exhibit weak-support rates of 9–14%, while cross-domain queries reach 60%. Ungrounded (fully unsupported) claims were not observed in this sweep; the dominant failure mode under the prior binary scheme was misclassification of weak support as strong support, not of unsupported as supported. We interpret the compositionality effect as consistent with broader queries requiring synthesis across a wider span of source material, yielding claims with weaker per-claim lexical and semantic alignment to any single retrieved passage.

7 Query Understanding

Ten distinct categories of query-parsing failure were identified. Nine were systematically closed, including interrogative and imperative phrasing, negation, relational phrasing (“the difference between X and Y”), qualified relational phrasing, Greek-letter and symbolic mathematical notation, typographical variants, and abbreviation/alias resolution. Each resolution is bound to a locked regression-test entry that fails under reintroduction of the corresponding defect, providing durable protection against silent regression. The tenth remains open and is discussed in Section 9.

8 Hardware Constraints and Model Selection

The system operates on an 8GB Apple M1 with the corpus resident on external disk. Model-capacity limits were established empirically rather than assumed, via direct stress-testing: a 14B-parameter instruction-tuned model was observed to silently truncate prompts beyond a 4,096-token context window irrespective of configured limits; an 8B-parameter model produced GPU out-of-memory failures above approximately 8,000 tokens of context. The operating envelope was fixed accordingly at approximately 7B parameters and 4K context, an empirically-grounded ceiling rather than a heuristic estimate.

Memory budget for the full verification stack, including the optional secondary embedder:

$$440 + 280 + 4,096 + 700 + 440 \approx 5,900\text{MB}$$

(retrieval embedder, reranker, 4-bit local LLM, vector index, secondary embedder, respectively, in MB) against

8,192MB of physical memory, the quantitative basis for keeping the secondary embedder disabled by default.

9 Limitations

Several open items are tracked rather than resolved, in the interest of scientific completeness. First, the secondary-embedder verification pathway (Section 4) is fully implemented but not yet validated end-to-end against the primary signal’s error modes, and currently runs disabled by default. Second, one query-parsing failure mode remains open, corresponding to the tenth category noted in Section 7: a trailing domain qualifier (e.g., “Banach spaces in functional analysis”) remains bound to an extracted concept following prefix-stripping, defeating literal-phrase anchoring; no general regular-expression-based resolution was identified, and the defect is deferred as low-incidence in practice. Third, repeat-run stability of claim-level tier assignments under stochastic generation has not yet been systematically characterized. Earlier development runs exhibited score variability, but those measurements predated the current abstention pathway and are therefore not reported as evidence for the present system; N -sample averaging remains the identified structural mitigation should this be characterized in future work. Fourth, retrieved document-ID sets exhibit mild drift across repeated trials of an identical query, consistent with embedder/reranker non-determinism on the local GPU compute backend; current test assertions tolerate this via any-match rather than exact-match criteria, an adequate but incomplete mitigation.

10 Future Work

The present system prioritizes verified retrieval and generation as a foundational layer. Planned extensions introduce a generative layer atop this foundation (including code synthesis grounded in retrieved paper content, with generated code verified against the passages it purports to implement), contingent on continued reliability improvements to the verification layer, under the governing premise that generation without a trustworthy verification substrate constitutes a regression relative to the present system, not an advance.

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